

Tutorial-5

1. Calculate the requested measures in parts (a) through (f) for bond A (assume that each bond pays interest semiannually):

(a) What is the price value of a basis point for bonds A?

	Bond A
Coupon	8%
Yield to maturity	8%
Maturity (years)	2
Par	\$100.00
Price	\$100.00

(b) Compute the Macaulay durations for the bond A.

	Bond A
Coupon	8%
Yield to maturity	8%
Maturity (years)	2
Par	\$100.00
Price	\$100.00

(c) Compute the modified duration for bond A.

	Bond A
Coupon	8%
Yield to maturity	8%
Maturity (years)	2
Par	\$100.00
Price	\$100.00

(d) Compute the approximate duration for bond A using the shortcut formula by changing yields by 20 basis points and compare your answers with those calculated in part (c).

	Bond A
Coupon	8%
Yield to maturity	8%
Maturity (years)	2
Par	\$100.00
Price	\$100.00

(e) Compute the approximate convexity measure for bonds A using the shortcut formula by changing yields by 20 basis points.

	Bond A
Coupon	8%
Yield to maturity	8%
Maturity (years)	2
Par	\$100.00
Price	\$100.00

2. State why you would agree or disagree with the following statement: If two bonds have the same dollar duration, yield, and price, their dollar price sensitivity will be the same for a given change in interest rates.

3. State why you would agree or disagree with the following statement: For a 1-basis point change in yield, the price value of a basis point is equal to the dollar duration.

4. Consider the following two Treasury securities:

Bond	Price	Modified duration (years)
A	\$100	6
B	\$80	7

Which bond will have the greater dollar price volatility for a 25-basis-point change in interest rates?

5. Consider the following portfolio:

Bond	Market Value	Duration (years)
W	\$13 million	2
X	\$27 million	7
Y	\$60 million	8
Z	\$40 million	14

(a) What is the portfolio's duration?

(b) If interest rates for all maturities change by 50 basis points, what is the approximate percentage change in the value of the portfolio?